

Aurora **Multi-Region** Architecture

Ensuring 99.99% Availability through Regional Affinity &
GoldenGate Replication



MANAGED VS. LOGICAL DEPLOYMENT

Metric	Aurora Global Database	Regional Affinity (GoldenGate)
Write Mode	Active-Passive (1 Primary Region)	Active-Active (Multi-Regional Writers)
Availability	Regional Failover Dependent	Local Survivability (Independent)
Conflict Risk	None (Single Writer)	Eliminated via Affinity Logic
Operational Ease	High (Native AWS Managed)	Medium (Requires GG Hub Mgmt)

CONTINUITY: THE "AFFINITY FLIP" LOGIC



Planned Maintenance

Scenario: Schema changes or DB Patching.

Action: Manually flip *ProgramAffinity* in DynamoDB. Traffic flows to Region 2 instantly. Database 1 patched with zero platform downtime.



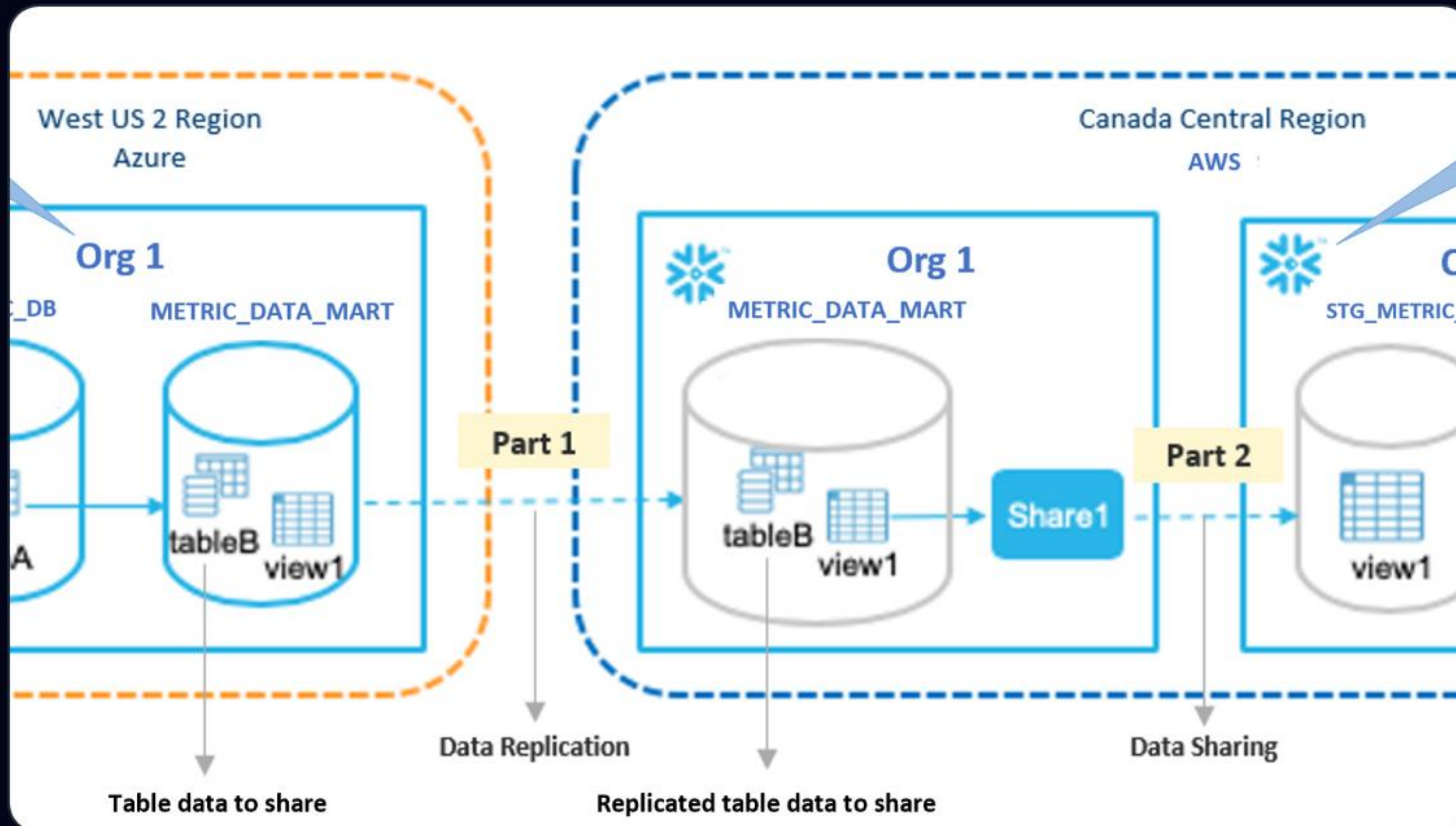
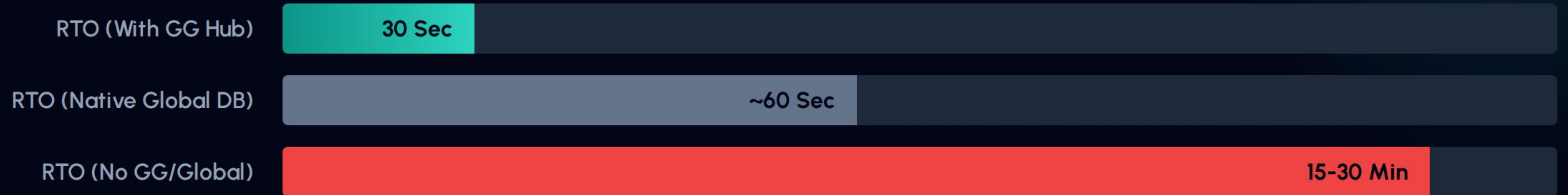
Unplanned Outages

Scenario: Region 1 becomes unavailable.

Action: Health Monitor Lambda detects loss, updates DynamoDB. All resolvers follow the *exact same* path as the planned maintenance.

Note: Using planned logic for unplanned outages ensures operational muscle memory and reliability.

BUSINESS IMPACT: RTO/RPO & AVAILABILITY



The Risk of Manual Recovery

Without GoldenGate or Global Database, business criticality is compromised. Manual failover involves DNS TTL delays, volume mounting, and manual consistency checks, ballooning RTO to 30+ minutes and risking multi-minute RPO loss.

REPLICATORS: NATIVE VS. GOLDENGATE

Capability	Aurora Native Logical	Oracle GoldenGate (GG)
Loopback Prevention	Manual (Requires origin filtering)	Built-in (TRANLOGOPTIONS)
DDL Replication	Not Supported (Manual Sync)	Fully Automated (Capture/Apply)
Conflict Resolution	None (App layer only)	Native CDR (Conflict Detection)
Transformations	Static / Per-table	High-flexibility Hub-based mapping

*Native logical replication lacks the "Enterprise Grade" features needed for complex bidirectional active-active environments.

CONFLICT PROTOCOL: PREVENTION & CDR

1. Prevention (Regional Affinity)

By assigning **ownership** of program P1 to Region 1, we ensure that writes never occur simultaneously in two places during the "Happy Path." This eliminates 99.9% of potential conflicts.

2. Resolution (GG CDR)

For the 0.1% "Split-Brain" scenario (failover moments), GG provides **CDR (Conflict Detection & Resolution)**:

- **USEMAX**: Last Update Wins (via Timestamp)
- **OVERWRITE**: Source wins unconditionally
- **DISCARD**: Prevents data corruption

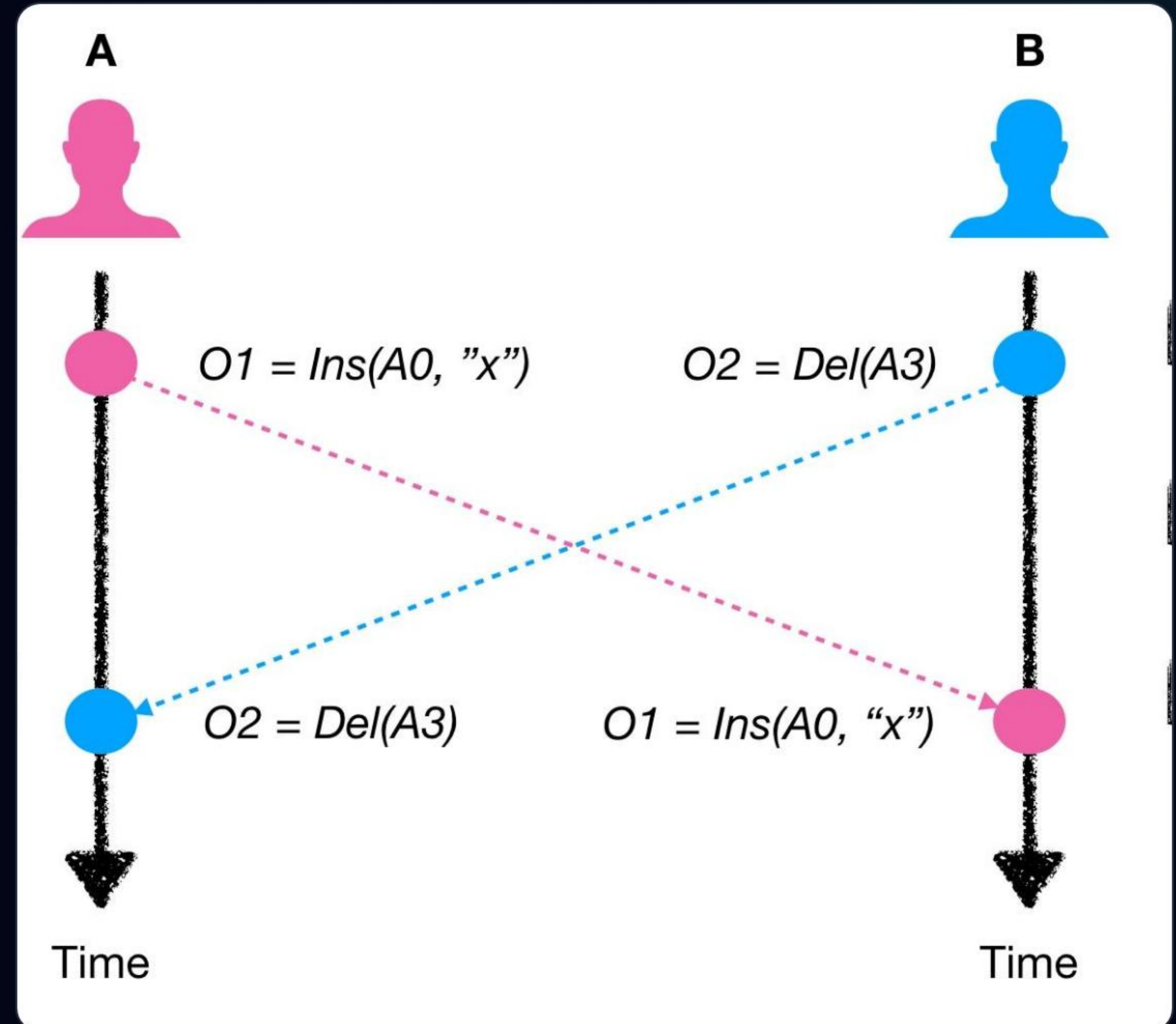
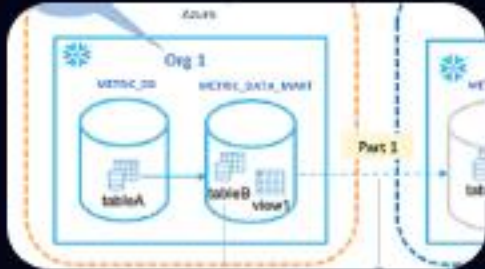


IMAGE SOURCES



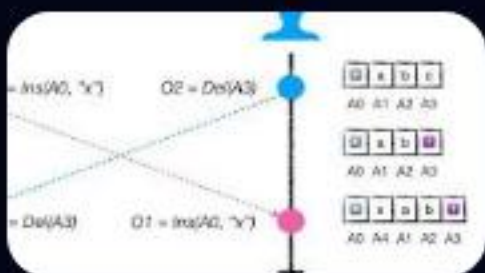
https://img.freepik.com/premium-vector/abstract-connection-dark-blue-background-hightech-network-technology-background-with-dots-lines-connection-system-vector-abstract-technology-blue-background_387612-692.jpg

Source: www.freepik.com



https://miro.medium.com/1*5tkyrBX60QUgDU9eGDLLTw.png

Source: medium.com



https://www.codecentric.de/_next/image?url=https%3A%2F%2Fcentral-1.graphassets.com%2FAiE4QoWSSilQO3k152ugkz%2Foutput%3Dformat%3Awebp%2Fdf16BOqdRkaWaFFg0jsk&w=3840&q=75

Source: www.codecentric.de